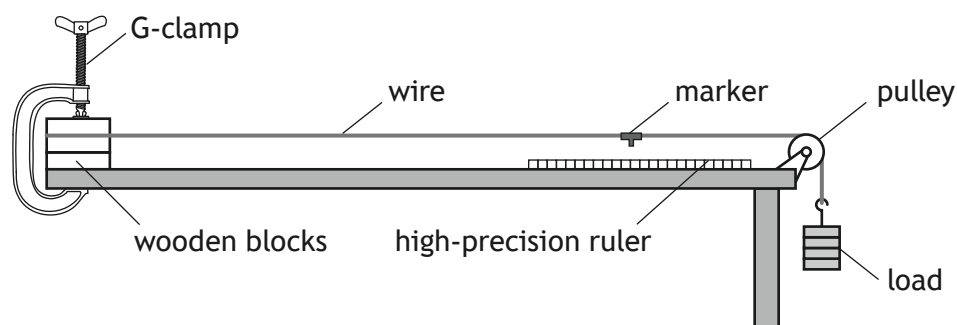


14. Young's modulus E of a material is a measure of its stiffness.
A student sets up an experiment to determine Young's modulus of a copper wire.



The student applies a force to the wire, which causes the wire to stretch.

The extension of the wire is the difference between its new length and its original length.

The student varies the force F applied to the wire and measures the corresponding extension e of the wire.

The student records the following information.

F (N)	e ($\times 10^{-3}$ m)
2.0	0.5
3.9	1.0
5.9	1.6
7.8	2.1
9.8	2.7
10.8	3.2

- (a) (i) Using the square-ruled paper on *page 48*, draw a graph of F against e . 3
(The table of results is also shown on *page 49*, opposite the square-ruled paper.)
- (ii) Calculate the gradient of the line of best fit on your graph. 2
Space for working and answer